Building Amazon Machine Images with HashiCorp Packer

HashiCorp Packer is an open source tool that helps to create identical machine images for multiple platforms from a single source template. This article shows how to use this tool to build Amazon machine images or AMIs.

Image Source: http://www.freepik.com/

Everyone knows that a virtual machine or a VM is not a physical machine. Instead, it is created using software that runs on a physical machine to emulate the functionality of another physical computer. The machine on which the VM runs is called a *host* and the machine it emulates is called a *target*. In other words, we can just launch a VM on a host to get the functionality of a target.

Many organisations have moved their functionality to VMs running on cloud infrastructure since it is hassle-free and works out cheaper. For example, being a popular cloud provider, AWS offers a facility to launch VMs on a vast number of their host machines. This service is called Elastic Compute Cloud or EC2 for short. You can build your entire computing infrastructure just by launching the required VMs on the AWS EC2 service.

To help you in building the VMs, AWS offers a few ready-to-be-launched target machine images or AMIs (AWS machine images). An AMI is a deployment unit that includes an operating system, utilities, and other resources. AWS offers AMIs to build Ubuntu VMs, Windows VMs, etc, to name a few. These are all general-purpose images.

In most cases, you may want to launch a custom VM for your special needs. Such a VM may be based on a specific version of an operating system, with a specific provisioning.

There are two ways of achieving this.

Mutable VM: In this approach, you launch the VM from a suitable AMI, install other utilities, and provision the ports and other resources. AWS offers a nice web interface to accomplish this. However, the problem with this approach is that you have to repeat the process for